

Review

The awakening of an advanced malignant cancer: An insult to the mitochondrial genome[☆]

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ABSTRACT

Background: In only months-to-years a primary cancer can progress to an advanced phenotype that is meta-static and resistant to clinical treatments. As early as the 1900s, it was discovered that the progression of a cancer to the advanced phenotype is often associated with a shift in the metabolic profile of the disease from a state of respiration to anaerobic fermentation — a phenomenon denoted as the Warburg Effect.

Scope of review: Reports in the literature strongly suggest that the Warburg Effect is generated as a response to a loss in the integrity of the sequence and/or copy number of the mitochondrial genome content within a cancer.

Major Conclusions: Multiple studies regarding the progression of cancer indicate that mutation, and/or, a flux in the copy number, of the mitochondrial genome content can support the early development of a cancer, until; the mutational load and/or the reduction-to-depletion of the copy number of the mitochondrial genome content induces the progression of the disease to an advanced phenotype.

General significance: Collectively, evidence has revealed that the human cell has incorporated the mitochondrial genome content into a cellular mechanism that, when pathologically actuated, can de(un)differentiate a cancer from the parental tissue of origin into an autonomous disease that disrupts the hierarchical structure-and-function of the human body. This article is part of a Special Issue entitled: Biochemistry of Mitochondria.

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1. Introduction

Respiration in a human cell is primarily localized to the mitochondria that reside in the cytosol of the eukaryote. The mitochondria harbor the only extra-nuclear genome in the form of a compact (~16,569 base pair) sequence of double-stranded DNA [1,2]. The two strands are differentiated by the nucleotide content with the guanine rich strand referred to as the heavy strand, and the cytosine rich strand referred to as the light strand. The heavy strand encodes 28 genes, and the light strand encodes 9 genes for a total of 37 genes. Of the 37 genes, 13 are for proteins (polypeptides), 22 are for transfer RNA (tRNA) and two are for the small and large subunits of ribosomal RNA (rRNA) [1,2]. It is important to note that the remaining polypeptides that are critical for the structure and the function of mitochondria are encoded in the nuclear genome. In a somatic cell the nuclear genome is represented as two copies of a linear sequence that is partitioned

into 23 chromosomes; on the other hand, the mitochondrial genome is a circular construct that is harbored at a copy number in the thousands within the mitochondrial matrixes of a single cell [3].

This review will begin with a general discussion of the mitochondrial genome, and the potential repercussions that are generated when a human inherits a variant (haplotype) of the genome. The review will, also, address the development and significance of somatic mutations in cancer, in addition to, the cellular effects that are produced as a response to a heteroplasmic to homoplasmic shift in the representation of a mutant sequence within the mitochondrial genome content. With respect to the copy number of the mitochondrial genome, the review will discuss the reports that strongly suggest that the early development of cancer is characterized with a flux in the copy number of the mitochondrial genome, until; a reduction-to-depletion of the copy number of the mitochondrial genome content is noted in the progression of the disease to an advanced phenotype. The review will then describe mitochondrial-genomic knock-out and mitochondrial genomic knock-in models which have confirmed that a loss in the integrity of the mitochondrial genome content of a cancer is transduced into the de(un)differentiation of the cells from the parental tissue of origin into an autonomous disease of enhanced metastasis, and resistance to the apoptotic effects of clinical therapeutics. In conclusion will be discussed the potential for the utilization of the mitochondrial genome

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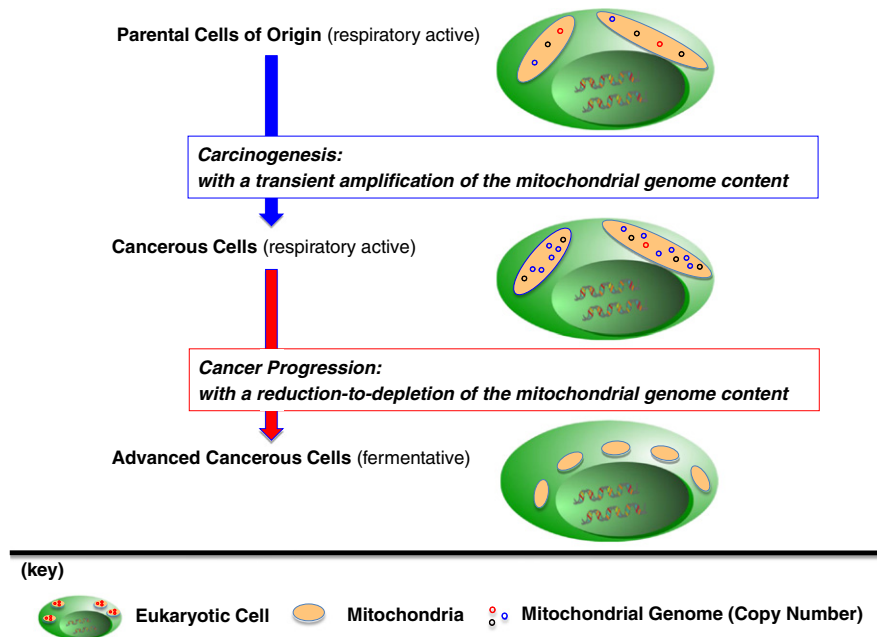


Fig. 1. The mitochondrial genome content in cancer development: In general, cells of human origin are respiratory-competent with mitochondria that harbor a mixture (haplotype) of sequences of the mitochondrial genome. Evidence indicates that in carcinogenesis the copy number of the mitochondrial genome is amplified, potentially as a consequence to the generation, and/or heteroplasmic-to-homoplasmic shift, of a mutant sequence of the mitochondrial genome; until, the progression of the cancer to an advanced (fermentative) phenotype is observed with a reduction-to-depletion of the mitochondrial genome content.

content as a biomarkers in clinical approaches to the maintenance and treatment of cancer and other degenerative diseases.

2. Maintenance of the mitochondrial genome

The transcription and translation of the 13 mitochondrial-encoded genes are mediated by an essential set of 22 tRNAs and 2 rRNAs (12S and 16S rRNAs) that are, also, encoded within the mitochondrial genome [1,2]. All 13 of the mitochondrial-encoded polypeptides are components of the mitochondrial respiratory chain (MRC), including: 7 of the at least 46 polypeptides in complex I, NADH dehydrogenase (ND1, 2, 3, 4L, 4, 5, 6); 1 of 11 polypeptides in complex III, cytochrome *bc*₁ (cytb); 3 of 13 polypeptides of complex IV, cytochrome c oxidase (COI, II, III); and, 2 of 16 polypeptides of complex V, *F*₁*F*₀ ATP

synthetase (ATP6 and 8) [1,2]. The remaining polypeptides of the MRC are nuclear-encoded genes; thus, the MRC is the only cellular structure to be encoded from the nuclear genome and the mitochondrial genome.

Within the mitochondrial genome is a non-coding displacement (D)-loop (~1122 base pairs) that harbors the main promoter for the transcription of the heavy strand and the light strand of the genome. Additional components of the D-loop include the origin of replication of the heavy strand, mitochondrial transcription factor (mtTFA) binding sites, and conserved sequence blocks (CSB I, II, III) [4]. The collective properties of the D-loop integrate nuclear-encoded processes into the maintenance, replication and transcription of the mitochondrial genome. Since the mitochondrial genome is expressed as polycistronic transcripts [5], it is important to note that the entire

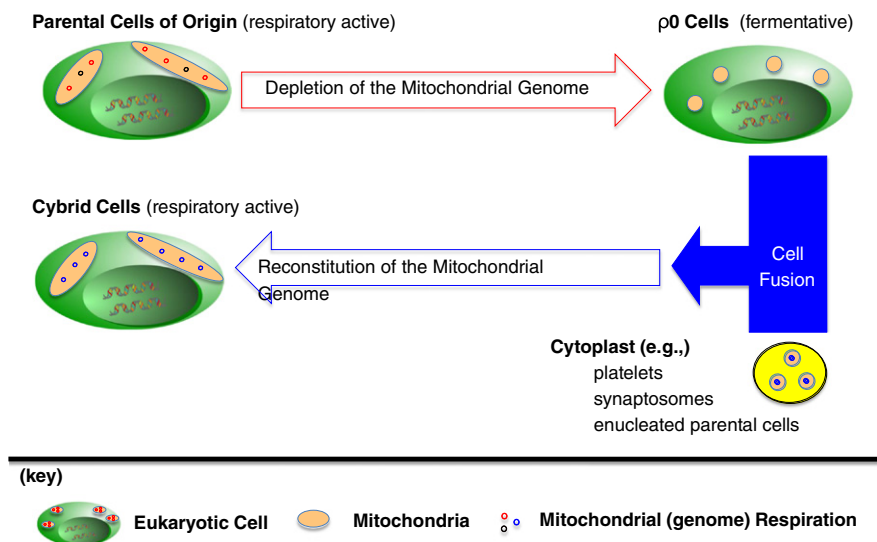


Fig. 2. Mitochondrial genome knock-in (p0) and mitochondrial genome knock-out (cybrid) models: In general, cells of human origin are respiratory-competent with mitochondria that harbor the mitochondrial genome. Viable cells (p0) can be generated with the depletion of the mitochondrial genome. As a control, the mitochondrial genome content can be reconstituted to p0 cells with the fusion of p0 to cytoplasm that act as membrane bound donors of mitochondria, and, thus, the mitochondrial genome.

repertoire of mitochondrial-encoded polypeptides can be potentially jeopardized with a mutation to the D-loop [6].

A reduction-to-depletion of the mitochondrial transcriptome can, also, be a consequence of mutation(s) to the nuclear genome [7–9]. Nuclear-encoded proteins that are essential for the replication of the mitochondrial genome include mitochondrial DNA polymerase γ (POLG), DNA helicase *twinkle*, and single-stranded DNA-binding protein (*mtSSB*). For the repair of the mitochondrial genome the mammalian mitochondria imports mechanisms, such as: elimination of mutagenic 8-oxodeoxyguanosine triphosphate (8-oxodGTP) [10], short patch base excision repair (SP-BER) [11,12], long-patch base excision repair (LP-BER) [13,14], mismatch repair [15,16], and non-homologous end-joining [17,18]. The nuclear-encoded maintenance of the mitochondrial genome, also, includes sanitation of the mitochondrial dNTP pool [19,20], and the selective destruction of damaged sequences of the mitochondrial genome [21,22].

The complexity of the cellular mechanisms that maintain the integrity of the mitochondrial genome is further evident with the degree of protection that can be generated from a single molecular component. For example, p53 is an enzyme that facilitates multiple facets of genomic maintenance and repair [23–25]. The mitochondrial genome content (sequence and copy number) is predisposed to somatic mutations in cells deficient of p53 [26,27]. New mechanisms for the maintenance of the mitochondrial genome continues to be discovered [28,29] and warrants a re-evaluation on the impact that a loss of integrity within the mitochondrial genome content generates in human health.

3. Mitochondrial genome haplotypes

The mitochondrial genome content is exclusively inherited from the oocyte in embryogenesis; thus, variations in the mitochondrial genome of a mother can be passed on to progeny as restriction fragment length polymorphisms (RFLPs). The isolation of subpopulations of humans over time, amongst diverse geographical locations on Earth, has generated a variety of polymorphisms within the mitochondrial genome. A polymorphism that has become established in a population is denoted as a haplotype. Within a human cell can be a combination of polymorphic sequences of the mitochondrial genome that are collectively classified as a haplogroup. The mitochondrial genomic haplogroups reflect major branch points in the mitochondrial phylogenetic tree and are named alphabetically (A to Z) in the order of discovery [30,31]. African descendants are reported to have the highest diversity of haplogroups in the world, including L1, L2, and L3 [32]. European and Asian decedents are often found to harbor the distinct set of haplogroups H, I, J, K, T, U, V, W, X and A, B, C, D, F, G respectively [33,34].

Maternal propagation of the mitochondrial genome can generate inherited diseases with the passage of mutations that shut down the electron transport (complex I–IV) and/or the ATP synthesis (complex V) of mitochondrial respiration [2]. In respect to the development of cancers, the inheritance of the mitochondrial genome has been determined to be a biomarker of carcinogenesis within an array of tissue, including; breast [35–37], prostate [38], thyroid [39], oral [40], endometrial [41,42], colorectal [43], and renal tissues [38,44]. Reports have suggested that the inheritance of haplogroup U in European descendants is associated with an increased risk of prostate and renal cancer in North American populations [38], however; other reports indicate that the risk of prostate cancer is not associated with the haplogroups of European [45–47] and Asian descent [48]. For example, the Carolina Breast Cancer Study has indicated that haplogroup U is a risk factor for breast cancer in African-American and North Indian women, but not European descendants [35,36,49]. The collective evidence suggests that the background of the nuclear-genome and/or environment of a cancer could modify the impact that its mitochondrial haplotype has in the development of the disease.

The biological significance of the inheritance of the mitochondrial genome has been verified with cybrid models that are generated from the complete trade of the mitochondrial genome contents between cells. Briefly, cybrid models can be constructed with the transfer of a mitochondrial genome content of an enucleated (donor) cell to a recipient cell that has been depleted of the mitochondrial genome. Cell lines depleted of the mitochondrial genome are denoted as rho (ρ 0) cells [50]. Mitochondrial genome contents of interest can, also, be evaluated with cybrid models that are generated from the fusion of ρ 0 cells with platelets or synaptosomes, i.e., the remnant cytoplasm of a megakaryocyte or a nerve terminal, respectively [51,52].

Kulawiec et al. has reported that when a ρ 0 cell was fused with platelets that contain a G10398A missense (ND3) polymorphism, from an African-American patient with breast cancer, the activity of NADH dehydrogenase was diminished, the generation of oxidative stress was amplified, and tumor growth was enhanced in comparison to a control that was constructed with the cell-fusion of the ρ 0 cells to platelets that were collected from healthy patients [51]. The experimental evidence supports observational analysis that has associated the G10398A polymorphism with an increased risk of cancer in Indian, African, Polish, and European descendants [35–37,49,53]. In general, the collective reports of haplotypes/groups in cancer development have revealed that inherited variants of the mitochondrial genome are key initiators in cancer development, and modifiers of carcinogenesis. Moreover, it has been postulated that specific mitochondrial haplotypes are differentially associated with etiological processes of carcinogenesis, e.g., HBV-hepatocellular carcinoma (HCC) versus alcohol-HCC [54].

4. Somatic mitochondrial mutations

Statistics from the American Cancer Society indicate that cancers, such as, breast and prostate cancer, are triggered to metastasize with the aging of a human [55]. A direct cause–consequence relationship has been associated with the aging of humans and the accumulation of somatic mutations in the mitochondrial genome [56,57]. The mitochondrial genome is predisposed to mutation as a consequence of the sheer number of replications of the genome in the life cycle of a cell, in addition to, the oxidative stress that is generated in the near vicinity of the mitochondrial genome content as a byproduct of mitochondrial respiration [58,59]. Mutations in the mitochondrial genome that are characterized with oxidative stress (e.g., purine G–A and T–C transitions) are often noted within cancers [60–62].

When mutation in the mitochondrial genome impairs the transport of electrons, or, the synthesis of ATP synthesis, within the MRC the dysfunction in mitochondrial respiration can further enhance the cellular level of oxidative stress [51,63]. Interestingly, cancer progression that is initiated with a reduction-to-depletion of the mitochondrial genome content is not always associated with a significant flux in super oxide, the source of reactive oxygen species (ROS) [64]. The evidence suggests that a loss in the integrity of the mitochondrial genome (sequence and/or copy number) can progress a cancer to an advanced phenotype by way of ROS-dependent and -independent events. For example, a persistent source of mitochondrial genomic damage can, also, be generated in response to mutant mitochondrial-encoded tRNA genes that form secondary structures that resemble the mitochondrial origins of replication [65,66].

The detection of temporal changes in a mitochondrial genome content with respect to the key developmental stages of a cancer has improved with advances in the ability to sequence the entire genome within a single-cancerous cell, adjacent benign cell, and circulating lymphocytes from a patient [3,54,67–69]. Evidence indicates that somatic mutations accumulate in the mitochondrial genome, in part, as a result of a lifetime of exposure to environmental factors such as ultraviolet light [70], bacteria, viruses [71], and tobacco [72,73]. For example, lung tissues from smokers are generally acknowledged to have a

mitochondrial genome content with an increased density of mutations, at a higher copy number [74], than is observed in tissues of non-smokers [72,73,75–77]. The environmental toxins can preferentially generate oxidative damage and/or form adducts in the mitochondrial genome over the nuclear genome [50,78–82].

The presence of a tumor may, in return, exert a field effect that leads to mutations in 'bystander cells' that are adjacent or distant to the tumor. The evidence indicates that cells directly exposed to the mutagens can generate cell-to-cell signals that, ultimately, generate in the anatomically distant 'bystander' cells a similar profile of acute or chronic genomic damage noted in the mitochondria of cells directly exposed to the mutagen [83–90]. The bystander effect appears to be a general phenomenon that is inducible by radiation [88,91], thermal injury [92,93], and chemo-therapeutics/toxicities [94,95]. Gorman et al. has demonstrated in an *ex vivo* system of human colorectal cancer that conditioned medium from an irradiated tumor can generate a rapid and transient increase in the frequency of mutant sequences of the mitochondrial genome in bystander tissue, as a secondary effect to dysfunction of mitochondrial respiration [88,91].

It is postulated that the D-loop of the mitochondrial genome is a mutational hotspot that bears the burden of mutations in the aging and carcinogenesis of human cells, often before the morphological indications of malignant transformation are visible [60,66,96–99]. For example, Tan et al. reported that tissues from smokers accumulate mutations more than those from nonsmokers and those tissues from nonsmokers accumulate mutations predominately in the D-loop, whereas mutations of smokers were observed in the D-loop and coding regions [73].

The postulate has been further supported by the evaluation of prostate needle biopsies from patients suspected of having prostate cancer. Patients that were found to be histologically and symptomatically benign contained somatic mutations that were confined to the D-loop. In patients with adenocarcinoma, especially of an invasiveness phenotype, mutations were randomly distributed across the mitochondrial-encoded genes [66,99]. The collective evidence strongly suggest that aging and carcinogenesis introduce mutations into the D-loop of the mitochondrial genome of a cancer, which can be followed by a subsequent burst of mutations throughout the genome as the disease progresses to an advanced phenotype [98,100–102].

5. Cancer development

The ability to evaluate if a loss in the integrity of the mitochondrial genome (sequence and/or copy number) generates or enhances the progression of a cancer is subject to the scrutiny of reverse causation. Since eukaryote cells are prone to rapid death after the acute inhibition of mitochondrial respiration, the issue of reverse causation cannot be resolved by pharmacological analysis alone. As an alternative, the mitochondrial genome content from patients or well characterized cell lines can be evaluated by cybrid models [103–108]. For example, cancers have been observed to progress to an advanced disease after obtaining a mitochondrial genome content from platelets of an African-American breast cancer patient with a polymorphic ND3 gene (G10398A) [51], or, cytoplasm generated from a MDA-MB-435 breast cancer cell line with a mutant tRNA gene (Leu(CUN)) [109]. The characterization of cybrid models strongly suggests that variants of the mitochondrial genome are modifiers in carcinogenesis, in addition to, primary initiators for the progression of cancer to an advanced disease [63,106,107].

Specific variants of the mitochondrial genome can differentially affect the function of the MRC; thus, it is critical in the study of cancer development that the biological significance of mitochondrial-encoded genes be evaluated on a gene-by-gene basis. Structural limitations hinder the single transfer of a mitochondrial-encoded gene into the mitochondria; and, unfortunately, the non-universal code of the mitochondrial-encoded genes limits the use of conventional expression vectors for the expression of mitochondrial-encoded

polypeptides. The direct translation of mitochondrial-encoded sequences in the cytosol results in multiple amino acid alterations and/or premature truncation. With advances in long-range gene synthesis, mutant mitochondrial-encoded genes have been isolated from primary cancer, and, subsequently, converted into nuclear (universal) codons that can be expressed from vectors in the cytosol. The vectors also allow for the addition of mitochondrial targeting sequences to the gene products. As a proper control, a wild-type of a specific mitochondrial-encoded gene can be converted to the universal-code that is, then, concurrently transfected with a variant of the gene. Such studies have shown that variants of ND2 (G4831A or A4605G) can enhance anchorage-dependent and anchorage-independent growth, in addition to, glycolysis in HeLa and head and neck squamous cancer [63]. Mutant ATP6 (T8993G or T9176C) has been observed to increase tumor growth and diminish the apoptotic sensitivity of HeLa cells [105]; while, a variation of the D-loop from colorectal cancer cells (SW480) is reported to enhance tumorigenicity [110]. The collective evidence further emphasizes the need for knowledge regarding how variants of the mitochondrial genome are biologically significant in the development of cancer.

6. Heteroplasmy to homoplasmy

A human cell has multiple copies of the mitochondrial genome, and, thus, a single cell can contain a variety of sequences of the genome. The coexistence of two or more sequences of the mitochondrial genome with slightly different nucleotide compositions in a single cell or tissue is defined as heteroplasmy [31]. In the early attempts of mitochondrial genomic analysis, experimental techniques were limited to the lysates of bulk tissues; and, heteroplasmic variations were noted to be quite rare in the mitochondria of healthy individuals. As quantitative-sequencing techniques improved to utilize the mitochondrial genomic content of single cells isolated from tissues, the sensitivity and resolution in the detection of variants of the mitochondrial genome were increased. Moreover, the advent of resequencing arrays allowed for evaluations of large quantities of DNA in a single run at low cost. The advances have revealed that heteroplasmy is observed and varies markedly in the mitochondrial genome content of a broad array of tissues within an individual [63,96,111–116].

It has been reported that in the progression of cancer (s) to an advanced disease a mitochondrial genome sequence can shift from a heteroplasmic-to-homoplasmic representation [66,117]. In a study of the carcinogenesis of normal colorectal mucosa, oncogenesis has been associated with the deletion of specific mutations in the cancer cells; while, a significant number of mutants are observed to become homoplasmic [66,96]. The heteroplasmic-to-homoplasmic shift of a variant sequence in the mitochondrial genome content of a cell can, also, be influenced by environmental factors as observed in buccal cells exposed to cigarette smoke [73].

Specific variants of the mitochondrial genome are known to be prone to undergo a heteroplasmic-to-homoplasmic shift. For example, Turner et al. has observed in a cybrid model that the introduction of the A3234G mutation (tRNA^{Leu}(UUR)) is subsequently followed by the mitotic segregation and homoplasmic representation of the mutation [118]. Multiple reports have demonstrated the biased segregation of additional variants of the mitochondrial genome, *in vitro* and *in vivo* [119–127]. If a variant of the mitochondrial genome produces a selective cellular advantage, over the thousands of generations required for tumorigenesis, the variant could become homoplasmic within a tissue as the oncogenic cell overtakes the tissue by clonal growth [128]. Alternatively, computer models suggest that during the unbiased states of replication and genomic sorting in the propagation of a cell, a variant sequence within the mitochondrial genome content can become homoplasmic [129]. Thus, warrants the need for the continued investigation to determine how the development of cancer is impacted as a response to shifts in the

proportions of variant sequences within the mitochondrial genome content of a cell.

7. Copy number, amplification to depletion (Fig. 1)

The mitochondrial genome content is reported to vary within the subpopulations of oncogenic cells collected from the same cancerous tumor. For example, Mizumachi et al. reported that tumors of prostate cancers are composed of cells that contain an amplified copy number of the mitochondrial genome content, in addition to, cells with a diminished-to-depleted copy number of the genome [3]. Mizumachi et al. also observed in primary prostate cancer cell lines a higher distributional variance in the copy number of the mitochondrial genome content than was reported in immortalized adjunct normal cells collected from the same tumor. It has been postulated that the copy number of the mitochondrial genome is amplified as a compensatory response to oxidative stress [58,130]. Unfortunately, an amplified copy number of the mitochondrial genome in a cancer could be an advantage for tumorigenesis, e.g., if a subsequent enhancement of mitochondrial-respiration enhances hypoxia-inducing signals and/or ROS-generated signals [131–134]. For example, in head and neck cancer it has been reported that the amplification of the mitochondrial genome content enhances the resistance of cells to docetaxel, in part, as a response to an increase in the activity of F_0 -ATPase [3].

While the copy-number of the mitochondrial genome of a cell is noted to be amplified after the carcinogenesis of lung tissue [135,136], the progression of the cancer to an advanced disease is often associated with a depletion of the mitochondrial genome and low oxidative stress [94]. In general, the depletion of the mitochondrial genome content is a common characteristic in the progression of a variety of cancers [102], as observed in ovarian [137], gastric [138], hepatocellular [6,139,140], renal cell [44,141–143], prostate [144–147], and breast cancers [147–150].

Multiple reports have shown that specific variants of the mitochondrial genome, particularly the hypervariability of the D-loop, are deleterious to the copy-number of the genome [6,61,149]. A complete depletion of the mitochondrial genome and/or the polycistronic transcriptome has been documented in cells following the heteroplasmic-to-homoplasmic shift of a mitochondrial genomic sequence(s) that harbors mutation mapping to a tRNA, rRNA or the D-loop [123,151–153]. For example, Turner et al. has demonstrated that a heteroplasmic-to-homoplasmic shift of the A3234G mutation (tRNA^{Leu}(UUR)) is frequently followed by a depletion of the mitochondrial genome [118].

From the perspective of nuclear-encoded genes, reports have shown that mutants of the primary machinery for the replication of the mitochondrial genome (e.g., DNA polymerase γ (POLG) and DNA helicase twinkle) can also perpetuate a dysregulation and depletion in the copy number of the genome [9,154,155]. For example, Singh et al. noted that the majority of breast cancers harbor a nuclear genome with mutant POLG genes, and mitochondria that are depleted of the mitochondrial genome; moreover, the report demonstrated that the expression of vector-encoded mutant POLG in breast cancer cell lines depleted the mitochondrial genome content, and signaled the progression of the cancers to enhanced states of invasiveness [8]. From another perspective, Pursell et al. has reported that the activity of POLG can be diminished if the level of ROS in a cell leads to a minute shift of the total cellular dGTP pool to 8-oxo-dGTP [156].

Damage to nuclear-encoded proteins (e.g., p53, Ras and p66shc) that perturb mitochondrial homeostasis or inter organelle signaling within a cell is likewise to contribute instability to the copy number of the mitochondrial genome [157–159]. Mutations in nuclear genes involved in deoxynucleotide metabolism (e.g., a p53-regulated subunit (p53R2) of ribonucleotide reductase, RNR) generate diseases that are characterized by a severe reduction in the mitochondrial genome

content [25,160–162]. Models of p53 null mice and p53 knockdown human primary fibroblasts have also been observed to have mitochondria that are depleted of the mitochondrial genome content [27]. It has been proposed that p53 associates with the mitochondrial genome through physical interactions within the genome [163], in addition to, modifying the functions of POLG [164] and mtTFA [165]. The repertoire of nuclear-encoded genes involved in the homeostasis of the mitochondrial genome continues to expand; recent evidence strongly implicates that the actomyosin cytoskeleton, in addition to, autocrine and paracrine signaling pathways (e.g., insulin-like growth factor, IGF-1, by way of mitochondrial pyrimidine nucleotide carrier 1, PNC1) are involved in the maintenance of the mammalian mitochondrial genome [28,166].

Exogenous toxins such as alcohol, benzene, and herpes simplex virus are also linked to a reduction-to-depletion in the copy number of the mitochondrial genome [167–170]. Exogenous toxins such as benzene, and herpes simplex virus are also linked to a reduction-to-depletion in the copy number of the mitochondrial genome, as reported in buccal, parotid and lung tissues [169,170]. Interestingly, in response to carcinogens, the mitochondrial genome content of a cell is often noted to be transiently amplified, and, then, depleted in a dose-dependent manner [58,73–77,130,135]. The endogenous formation of toxins, such as, abnormalities in endocrine (estrogen and androgen) signaling pathways that develop in the progression of cancers, has also been associated with a reduction in the copy number of the mitochondrial genome [171–173].

8. Mitochondrial genomic knock-out ($\rho 0$) and mitochondrial genomic knock-in (cybrid) models (Fig. 2)

The mitochondria as a whole cannot be excluded from cells, as the mitochondria serve multiple roles for the eukaryote, including ATP synthesis, redox regulation, thermogenesis, calcium regulation, and production of secondary messengers [2]. From the perspective of the mitochondrial genome, the generation of viable ($\rho 0$) cell lines after the depletion of the genome from cancer cell lines reveals that a loss in the integrity of the mitochondrial genome content (sequence and/or copy number) progresses the disease to an advanced phenotype [145–147,150,174]. It should be noted that the in vitro cultures of $\rho 0$ cells are auxotrophic for pyruvate and uridine [50,175]. The observations can be attributed to the fact that uridine synthesis requires mitochondrial respiration, and that the ability to sustain a persistent flux in glycolysis is dependent on the conversion of pyruvate to lactate as a means to oxidize NADH to NAD^+ . In terms of the in vivo growth of $\rho 0$ cells, multiple reports have demonstrated the ability to generate xenograft tumors of $\rho 0$ cells in immune-deficient mice [176,177].

Within human cells that are competent for mitochondrial respiration the mitochondria form an elaborate tubular network of mitochondrial respiration that maintains the mitochondrial membrane potential. The mitochondria of the $\rho 0$ cells appear as punctate spots that localize around the nucleus [178]. Interestingly, the mitochondria of (respiratory-incompetent) $\rho 0$ cells can still obtain nuclear encoded proteins [179,180], a process that is generally acknowledged to be mediated by the mitochondrial membrane potential [181]. Evidence indicates that $\rho 0$ cells generate the mitochondrial membrane potential by the remnant components of F_1F_0 ATP synthase; it has been postulated that the mitochondrial membrane potential is generated, in part, as a response to an import of 'glycolytic' ATP by way of ATP synthase, that is coupled, to the (reverse) ATP/ADP translocase activity of ANT [179,182]. Reports also suggest that the mitochondrial membrane potential is maintained with additional electrogenic pumps, e.g., the mitochondrial chloride intracellular channel (mtCLIC) [179,183].

It is possible that in the formation of $\rho 0$ cells a mutation can be incidentally introduced into the nuclear genome that, ultimately, mask

the cellular signaling pathways and phenotypes that are generated with the depletion of the mitochondrial genome content. As a proper control to a $\rho 0$ cell line, a mitochondrial genome content can be reconstituted to $\rho 0$ cells with the creation of cybrid cell line, i.e., the fusion of a $\rho 0$ cell to cytoplasmic vessels that harbor donor mitochondria from platelets, synaptosomes or enucleated parental cells of the $\rho 0$ [51,52,146]. The shared nuclear background in a $\rho 0$ /cybrid model allows for the verification that a cellular event or phenotype which is observed in the $\rho 0$ and not the cybrid is a direct result of the depletion of the mitochondrial genome, and not the product of an incidental mutation to the nuclear genome.

9. A cellular and pathological signaling pathway

The $\rho 0$ cells are characterized with a reduced sensitivity to death-stimuli in comparison to parental cells. For example, Higuchi et al. reported that the depletion of the mitochondrial genome content in ML-1a cells generated a resistant to TNF-induced apoptosis, even though TNF-induced cellular proliferation and differentiation could still be observed in the $\rho 0$ cells [174]. The report also noted that the mitochondrial apoptotic machinery in the $\rho 0$ cells was not completely obliterated, based on evidence that the viability of the culture of the cells was decreased in the presence of staurosporine-induced apoptosis [174]. Suzuki et al. would later reveal that the $\rho 0$ cells generate a resistance to TNF-induced apoptosis by the constitutive activation of Akt [184]. A flux in the Akt signal is a key mechanism that can enhance the resistance of cancer to apoptotic-stimuli [51,145,185].

Additional reports of the molecular pathways within $\rho 0$ cells have demonstrated that Akt is one of a diverse array of signals (e.g., Raf/MAPK, PI3K/Akt, JNK, AP1, and NF- κ B) that are concurrently and reversibly activated with flux in the copy number of the mitochondrial genome content within a cell [64,147,184]. For example, Naito et al. noted that the depletion of the mitochondrial genome in breast cancer and prostate cancer generated the constitutive activation of the Raf/MAPK signaling cascade, in addition to, the enhanced expression of transforming growth factor beta (TGF β) 1 and type I TGF β receptor (TGF β RI) [147]. Naito et al. also observed the $\rho 0$ cells of the cancers to be of a mesenchymal-like and invasive phenotype, in comparison to the epithelial-like phenotype of the parental cells. In a corresponding report, Xie et al. noted in $\rho 0$ cells that the depletion of the mitochondrial genome generated the hypermethylation of promoter regions (i.e., DNA methyltransferase 1), and, thus, the silencing of epithelial biomarkers (E-cadherin) and a variety of tumor suppressor genes [186]. It is also heavily reported that the depletion of the mitochondrial genome content in $\rho 0$ cells elevates the transcription of genes relevant to metal homeostasis, initiation of EMT, glucuronidation pathways, autophagy of defective mitochondria, and elimination of lipophilic molecules via peroxisomal lipid metabolic pathways [187–189]. Furthermore, the silencing of systemic (hormone) signaling pathways that maintain the viability, growth, and differentiation of prostate cancer and breast cancer has also been noted in $\rho 0$ cells [144,146–149]. The overall phenotype of the $\rho 0$ cells strongly indicates that a depletion of the copy number of the mitochondrial genome in cancer is integrated into a mitochondrial-generated progression signal that generates the progression of the disease to an advanced phenotype that is highly invasive-to-metastatic, and resistance to clinical (apoptotic) stimuli [145–147,150,174].

10. An intracellular and extracellular biomarker

With the high frequency of variant sequences of the mitochondrial genome in cancer, at a copy number that is often more numerous than any mutant nuclear-encoded gene, the mitochondrial genome is potentially a convenient biomarker for cancer development. Reports have shown that the copy number of the mitochondrial genome is a biomarker associated with the risk of carcinogenesis in tissues

such as the lung [135,136]. Furthermore, mitochondrial genomic analysis can potentially enhance the histopathological evaluation of tumors and could be beneficial in the classification of malignant, benign, and proximal to malignant lesions [190]. Evidence also suggests that the characterization of the mitochondrial genome content (sequence and copy number) within cells could aid the identification of the etiology behind the carcinogenesis of the tissue [54], the designation of effective therapeutic regimens [3,108,146,150], and the prognosis for a patient [191].

Outside of the cell, fractions of the mitochondria genome of cancers have been isolated from extracellular fluids such as saliva [192], blood and plasma [97,193], urine [97,112], pancreatic juice [112], breast nipple aspirate fluid [194], and cerebral spinal fluid [195]. It is important to note that the measurement of circulating fragments of the mitochondrial genome can be masked by paraneoplastic characteristics; for example, enhanced protease activity can cause a decrease in the nucleic acid binding capacity on the surface of blood cells (erythrocytes and leukocytes) [193,196–198], nuclear complexes [199], and/or apoptotic bodies [200].

Accumulation of fragments of the mitochondrial genome (DNA, mtDNA) or transcriptome (RNA, mtRNA) in extra-cellular fluids is presumably generated by, a active release of the molecules from circulating cancer cells [201–203], in addition to, a passive release in response to apoptosis or necrosis of cancer and bystander tissue [204–207]. The quantification of circulating mtDNA has been postulated to be a prognostic marker of cancers that is of an increased sensitivity and specificity over circulating nuclear DNA [191]. It has been reported that a significant increase of mtDNA is observed in the saliva of patients with head and neck cancer, and, the cerebral spinal fluid of patients with medulloblastomas, in comparison to patients with no cancer [192,195]. The circulating mtDNA levels were also observed to decrease after radiotherapy or hormone therapy. A similar observation has been characterized for prostate cancer. Patients that die from prostate cancer within 2 years of initial presentation are reported to maintain higher levels of mtDNA in circulation, in comparison to patients with prostate cancer who survived for longer periods [191,208]. A concurrent elevation of circulating mtDNA and PSA has also been characterized with the early biochemical recurrence of prostate cancer after radical prostatectomy [191]. The collective observations suggest that intracellular and extracellular levels of the mitochondrial genome can be in direct relationship with the cancer cell burden, and, thus, provide a biomarker for the disease-free and the overall survival status of a patient diagnosed with cancer.

11. Revival of the mitochondrial genome

It is generally acknowledged that the endosymbiotic relationship of the mitochondrial genome and nuclear genome in the eukaryote enables the human cell to form tissue structures that can support the complex functions of the human body. An appreciation of the significance of the mitochondrial genome in cancer development was initially delayed with, the late discovery of the mitochondrial genome in comparison to the nuclear genome, in addition to, a time lapse in the adaptation of nuclear genomic analysis to the unique characteristics (sequence and copy number) of the mitochondrial genome content of a cell. Meanwhile, the identification of oncogenes and tumor suppressors revealed a broad spectrum of cellular signaling pathways that can, switch the cellular proteome from a metabolic state of mitochondrial respiration towards glycolytic fermentation, and, concurrently, de(un)differentiate a cancer into an advanced disease [209–211]. Near half of a century has passed since the discovery of oncogenes and tumor suppressors and cancer progression still remains a deadly event in human society [55,212]; thus, a return is warranted to the classical observations of the Warburg Effect that hinted to the postulate that the mitochondrial genome is more than a compensatory mechanism to genomic insults [213,214].

This review presented multiple lines of evidence which demonstrate that a loss in the integrity of the mitochondrial genome can lead to the de(un)differentiation of a variety of cancers from the parental tissue of origin, into autonomous (fermentative) diseases that disrupt the organization of tissue structure and function [146,147,150]. Interestingly, the depletion of the mitochondrial genome from cancer generates a de(un)differentiated cell that harbors mitochondria of similar morphology and genealogy as observed in undifferentiated embryonic stem cells (ESCs) and pluripotent blastomeres (pre-implantation embryos) [178,215]; the small and immature mitochondria are localized around the nucleus, and have a low copy number of the mitochondria genome and diminished MRC activity. From the alternative perspective, mitochondria in respiratory-competent cancers resemble eukaryotes that have differentiated to a specific cell fate [178,215]; the mitochondria are observed to form dense and wide networks with enriched mitochondrial genome contents and enhanced MRC activity.

The collective evidence suggests that the early (fermentative) stages of embryogenesis are maintained by a mitochondrial-generated progression signal that is generated in response to a low copy number of the mitochondrial genome. Moreover, the evidence strongly indicates that the mitochondrial-generated progression signal is pathologically 'reawakened' in the later stages of human life when the copy number of the genome is reduced within an adult (respiratory competent) cell. To this date no study has elicited a mechanism that links the mitochondrial genome content to a mitochondrial-generated progression signal of embryogenesis, or the progression of cancer. It is plausible that a loss in the integrity of the mitochondrial genome is a common pathological event that drives the progression of a broad spectrum of human diseases that are attributed to respiratory-incompetent mitochondria, such as the clearly defined 'mtDNA depletion syndrome' [216–219]. Future studies are warranted to define the mechanism(s) that transduces the integrity of the mitochondrial genome content into a mitochondrial-generated progression signal; such a discovery could reveal cellular processes that could be targeted in the attempt to stall and/or reverse the progression of cancer and other degenerative diseases, to enable a realistic fight as we await the cure.

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